

1 Assignment 0

1.1 Setup

Table 1: Machine and Software Configuration

Component	Specification
CPU Model	Intel(R) Core(TM) i5-10310U CPU @ 1.70GHz
Physical Cores	4
Logical Processors / Threads	8
RAM Amount	23.61 GB
SSD Amount	476.94 GB
OS Version	10.0.26200 N/A Build 26200
PostgreSQL Engine Version	PostgreSQL 18.6

1.2 Database

Name: Join Order Benchmark (2013 Snapshot)

This database is based on the 2013 snapshot. The database contains 21 tables.

Table 2: JOB Database Tables, Primary Keys and Cardinalities

Table Name	Primary Key	Cardinality
aka_name	id	901,343
aka_title	id	361,472
cast_info	id	36,244,344
char_name	id	3,140,339
comp_cast_type	id	4
company_name	id	234,997
company_type	id	4
complete_cast	id	135,086
info_type	id	113
keyword	id	134,170
kind_type	id	7
link_type	id	18
movie_companies	id	2,609,129
movie_info	id	14,835,720
movie_info_idx	id	1,380,035
movie_keyword	id	4,523,930
movie_link	id	29,997
name	id	4,167,491
person_info	id	2,963,664
role_type	id	12
title	id	2,528,312

2 Assignment 1

PostgreSQL maintains column statistics after the `ANALYZE` command. For this question, the histograms for the `id` and `title` columns of the `title` table were examined.

2.1 Question 1

2.1.1 1. Corresponding SQL

The statistics were first generated using:

```
ANALYZE title;
```

The histogram information was then obtained from the PostgreSQL `pg_stats` view using:

```
SELECT
    attname,
    n_distinct,
    most_common_vals,
    most_common_freqs,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
    AND attname IN ('id', 'title');
```

The `histogram_bounds` column contains the values used as histogram boundaries. PostgreSQL exposes these statistics through `pg_stats`; the underlying statistics are produced by `ANALYZE`.

2.1.2 2. Type of Histogram

The histogram maintained by PostgreSQL for these columns is an **equi-depth (equal-frequency) histogram**.

The histogram boundaries divide the values into ranges containing approximately the same number of tuples. Therefore, the widths of the value ranges need not be equal.

For example, if a histogram contains B buckets, each bucket approximately represents

$$\frac{1}{B}$$

of the tuples, subject to the effect of most-common values and the sampling used by `ANALYZE`.

2.1.3 3. Number of Buckets

The number of histogram buckets is one less than the number of entries in the `histogram_bounds` array.

For my PostgreSQL statistics:

- `id`: 100
- `title`: 100

The number can be obtained from PostgreSQL using:

```
SELECT
    attname,
    cardinality(histogram_bounds) - 1 AS number_of_buckets
FROM pg_stats
WHERE tablename = 'title'
AND attname IN ('id', 'title');
```

2.1.4 4. How the Textual Attribute title is Put into the Histogram

The `title` attribute is a textual value, so it cannot be treated as a numeric value directly.

PostgreSQL first orders the text values according to the column's comparison rules and stores selected ordered values as histogram boundaries. These boundaries divide the ordered text domain into approximately equal-frequency ranges.

For a range predicate on a text column, PostgreSQL converts the text value and the surrounding histogram boundaries to a common scalar comparison scale. It then performs interpolation inside the corresponding histogram bucket.

For strings, PostgreSQL uses the byte representation of the strings as a fractional scalar representation. Common prefixes are removed before the interpolation calculation so that the comparison can focus on the portion of the strings that differs.

Thus, a text histogram does not divide strings according to a numerical magnitude. Instead, it uses the ordering of the strings and a scalar representation suitable for interpolation.

2.1.5 5. Histogram Bucket Boundaries

The bucket boundaries obtained from `pg_stats` are listed below.

`id` histogram

```
{
    58, 24549, 51112, 77316, 100354, 127446, 152219, 175234,
    199351, 226185, 252917, 274611, 298665, 323475, 348752,
    373432, 399278, 422974, 447102, 472333, 496158, 520173,
```

```

546680, 576310, 602304, 628338, 653665, 681361,
707128, 729809, 756258, 782201, 807417, 835118, 863206,
887242, 911779, 936011, 960911, 985388, 1011785, 1035950,
1058604, 1086417, 1112274, 1136653,
1160531, 1186749, 1211107, 1235379, 1259253, 1281711,
1305714, 1330264, 1356923, 1381561, 1405457, 1431660,
1455036, 1481513, 1505512, 1534241, 1559580, 1587278,
1611445, 1637249, 1665180, 1690643, 1715930, 1740280,
1766040, 1789855, 1812673,
1837244, 1864195, 1889430, 1914964, 1939219, 1964987,
1991435, 2015811, 2042317, 2069297, 2095340, 2121752,
2148408, 2173353, 2197551, 2222369, 2247242, 2274114,
2303748, 2328246, 2354510, 2379115, 2403508, 2430141,
2456620, 2480455, 2505196, 2528302
}

```

For example, the boundaries should be interpreted as

$$[b_0, b_1, b_2, \dots, b_B]$$

where bucket i covers the ordered range

$$[b_{i-1}, b_i].$$

title histogram

```

{
  "'A' de abuelo",
  (#1.145), (#1.2686), (#1.4692), (#1.7554),
  (#13.10), (#2.45), (#30.6), (#50.84), (#9.124),
  (1974-03-15), (1989-07-11), (1996-06-14),
  (1999-05-31), (2001-03-21), (2002-12-09),
  (2004-07-16), (2005-11-04), (2007-02-22),
  (2008-06-11), (2010-03-01), (2011-05-21),
  (2012-06-14), (2013-05-16),

  "A Better Tomorrow",
  "A Sculptor's Landscape",
  "Aflevering 421",
  "ALS Video #39: Jamaica I Shoot - Part 2",
  "Anonymous Letters",
  "Auf dem Teppich",
  "Barry Bracelet Boffo",
  "Beware of the Oglethorpe",
  "Bob's Videos 103: Soft & Sexy",
  "Budva na pjenu od mora",
  "Carlos Jr.",

```

"Child Support Assurance",
 Collecting/Chores,
 "Crashus Maximus",
 "Dark Age of Light: Episode 01",
 "Death Tide",
 "Descubra México: Quintana Roo",
 Directions,
 "Dream No Evil",
 "El abrazo de los peces",
 "Enter Another Dragon",
 Faces,
 Finale,
 "Frances and Frankie",
 "Gay Cuba",
 "Going Places",
 "Haha no tomoshibi chiisaki mono wo terashite:
 Ishii fudeko chiteki shōgai ji kyōiku no michi",
 Hennè,
 Homespun,
 "I Have a Stomachache.",
 "In the Night Garden...",
 "J2: Rise of the Dragons",
 "Josh and Cindy's Wedding Trip",
 "Ke arteko egunak",
 "Kuan yin",
 "La maestrina",
 "Le barbier de Séville",
 "Lexi Gets Stiffed",
 "Los Elegidos",
 "Magic Hat",
 "Massacres: New Poetry for Remembering",
 "Miguel Returns to Ella",
 "Motown Maurice Hustle & Shuffle",
 "Nakuna inunaki trio",
 "Nis, Radojkovic",
 Ochtendverkeer,
 "Operation Makeshift",
 "Parfum troublant",
 "Pierde a los dos",
 "Power Rangers Time Force: The End of Time",
 "Qué lolas",
 Rejection,
 "Roinilan talossa",
 "San Mateo, CA",
 "Sehnsucht nach Liebe",
 "Show Forty Five",

```

Socialdemokratiet,
Stacja,
"Summer Tour: MITP Performance 1",
"Tango of Togetherness",
"The American Soldier in Love and War: Number 2",
"The Case of the Golden Fraud",
"The Emperor's Nightingale",
"The Ground We Stand On",
"The Legend of Robin Hood",
"The New Neighbors",
"The Rent Racket: Part 2",
"The Tale of a Feather",
"Them Was the Happy Days!",
"Tödliches Viereck",
"True Cost of Coal",
"Una nevicata a Milano",
"Vendetta I",
"Warsaw Rebuilds",
"Who Am I? Asian Pacific and Proud",
"Xtreme Fakeover",
"Zywot czlowieka rozbrojonego"
}

```

The values are ordered lexicographically according to PostgreSQL's comparison rules for the `title` data type.

2.1.6 6. Histogram Bar Plots

The histograms were visualized using the bucket boundaries on the X-axis and the corresponding frequency on the Y-axis, as required in the assignment.

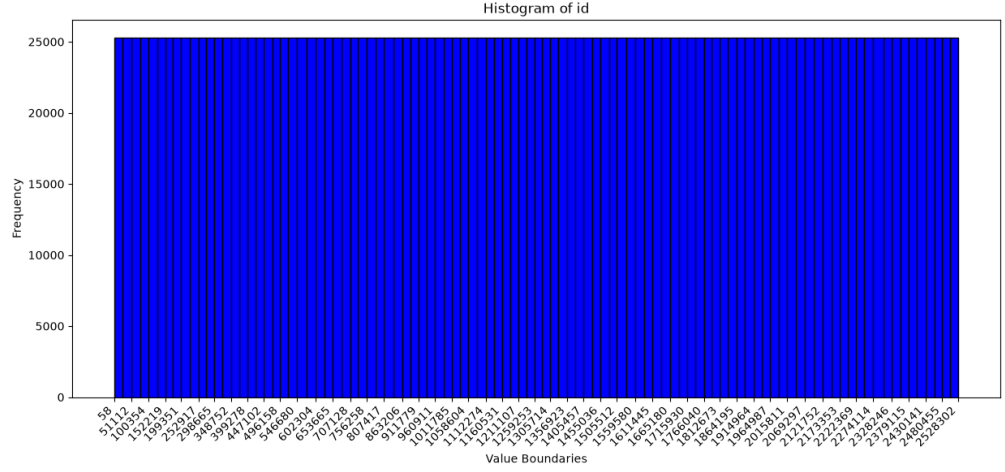


Figure 1: PostgreSQL histogram for `title.id`.

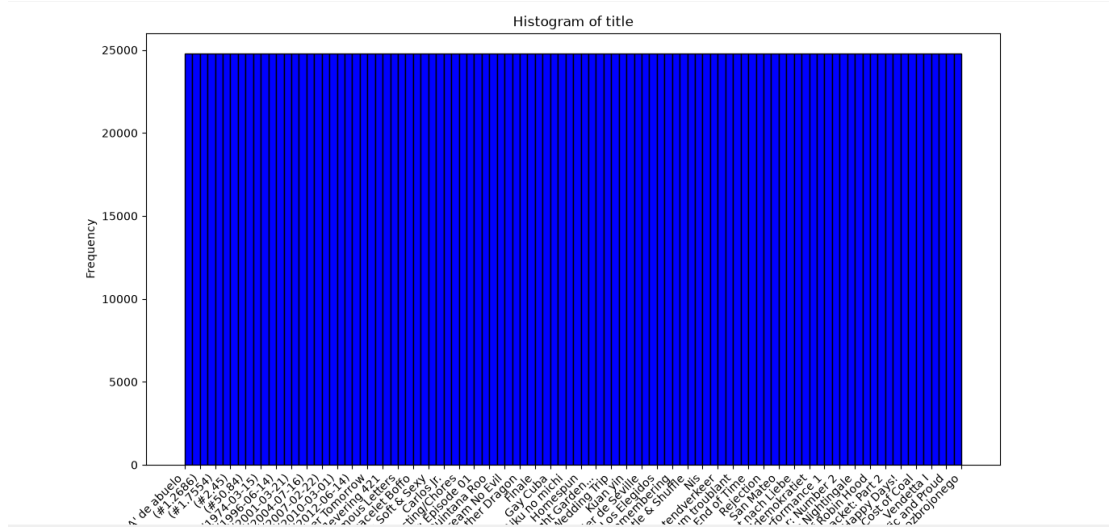


Figure 2: PostgreSQL histogram for `title.title`.

2.1.7 7. Maximum Selectivity Error

For a histogram bucket, PostgreSQL assumes an approximately uniform distribution of values within that bucket.

For a query value q falling inside a bucket with lower and upper boundaries L and U , the estimated fraction inside the bucket is

$$f = \frac{q - L}{U - L}.$$

If the bucket contains an equal fraction of the tuples, the estimated selectivity is obtained by adding all complete preceding buckets and the interpolated fraction of the current bucket.

For a histogram with B buckets, if q lies in bucket i , the estimated selectivity can be represented as

$$\hat{S}(q) = \frac{i - 1 + f}{B}.$$

The actual selectivity is calculated directly from the table:

$$S(q) = \frac{|\{t \mid t.A < q\}|}{|R|}.$$

Therefore, the absolute selectivity error is

$$E(q) = \left| \hat{S}(q) - S(q) \right|.$$

The maximum selectivity error is

$$E_{\max} = \max_q E(q).$$

For my results:

- **id:**

$$E_{\max} = \mathbf{0.01}$$

- **title:**

$$E_{\max} = \mathbf{0.01}$$

The error was calculated by comparing the histogram-based estimate against the actual selectivity obtained from the complete **title** table.

2.1.8 8. Selectivity Estimation for a Text Range Query

For a range query such as

`WHERE title < 'Race'`

PostgreSQL first locates the histogram bucket containing the value **'Race'**.

It then converts the query value and the two surrounding text boundaries to a common scalar scale. If the query value is between the two boundaries, PostgreSQL assumes an approximately uniform distribution inside that bucket and linearly interpolates the fraction of the bucket covered by the predicate.

If the query value lies before the first boundary or after the last boundary, the corresponding edge case is handled directly.

Thus, the selectivity is approximately

complete preceding buckets + fraction of the current bucket.

This allows PostgreSQL to estimate range predicates even though `title` is a textual attribute.

2.1.9 9. Estimated and Actual Selectivities

Query 1: `title < 'Race'`

The query is:

```
SELECT *
FROM title
WHERE title < 'Race';
```

The estimated number of rows can be obtained using:

```
EXPLAIN
SELECT *
FROM title
WHERE title < 'Race';
```

The actual number of rows is obtained using:

```
SELECT COUNT(*)
FROM title
WHERE title < 'Race';
```

Let

$$N = 2,528,312$$

be the cardinality of the `title` table.

Then:

$$S_{\text{estimated}} = \frac{\text{estimated rows}}{N}$$

and

$$S_{\text{actual}} = \frac{\text{actual rows}}{N}.$$

My results are:

Quantity	Value
Estimated rows	1,938,726
Actual rows	1,919,892
Estimated selectivity	0.7668
Actual selectivity	0.7594
Absolute error	0.0074

Query 2: `id < 350000`

The query is:

```
SELECT *
FROM title
WHERE id < 350000;
```

The estimated number of rows is obtained using:

```
EXPLAIN
SELECT *
FROM title
WHERE id < 350000;
```

The actual number of rows is obtained using:

```
SELECT COUNT(*)
FROM title
WHERE id < 350000;
```

The selectivities are calculated as

$$S_{\text{estimated}} = \frac{\text{estimated rows}}{2,528,312}$$

and

$$S_{\text{actual}} = \frac{\text{actual rows}}{2,528,312}.$$

My results are:

Quantity	Value
Estimated rows	355245
Actual rows	349999
Estimated selectivity	0.1405
Actual selectivity	0.1384
Absolute error	0.0021

2.2 Q2. Optimal Serial Histogram

For this experiment, an optimal serial histogram was constructed using 20 buckets. The histogram minimizes the objective

$$\sum_i n_i V_i,$$

where n_i is the number of distinct values in bucket i and V_i is the variance of the frequencies within that bucket.

The histograms were constructed for the `id` and `title` attributes of the `title` table.

2.2.1 Q2.1 – Histograms for 3000 Samples

The requested sample size was 3000 rows. Due to the use of `TABLESAMPLE`, the actual number of sampled rows was 2809.

The resulting optimal serial histograms contain 20 buckets.

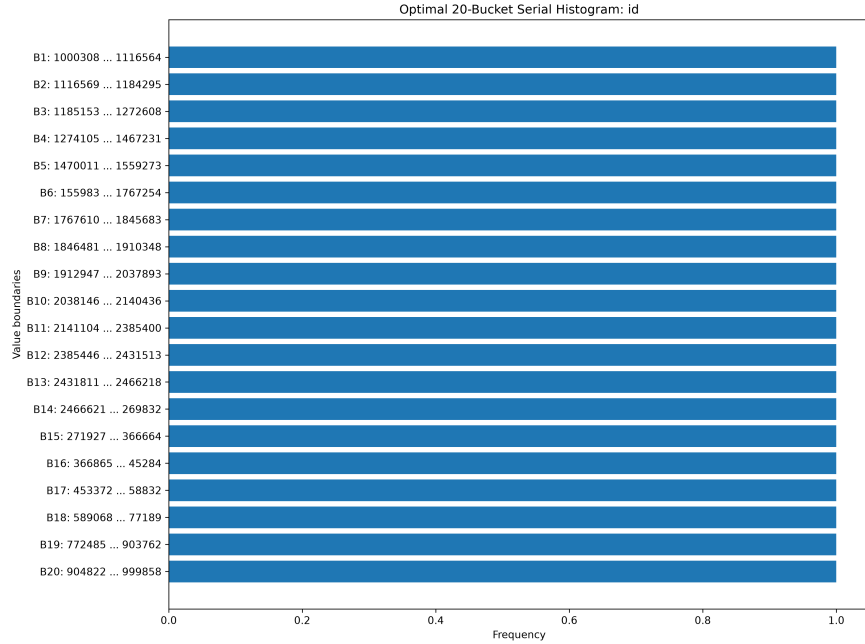


Figure 3: Optimal 20-bucket serial histogram for `id` using the 3000-row requested sample.

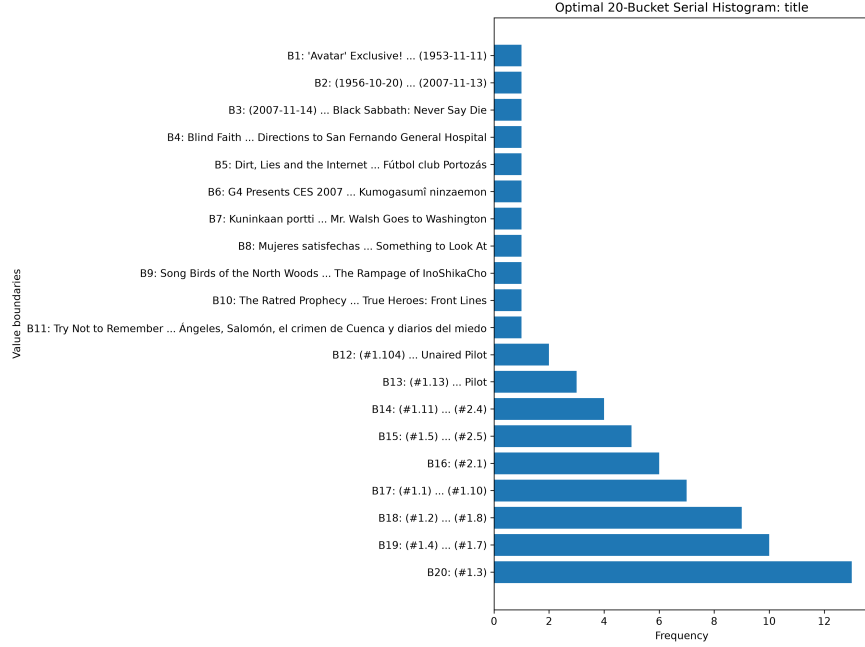


Figure 4: Optimal 20-bucket serial histogram for `title` using the 3000-row requested sample.

For the `id` attribute, all sampled values occur with frequency 1. Consequently, every bucket has mean frequency 1 and zero variance.

For the `title` attribute, most sampled titles occur once, while a smaller number of titles have higher frequencies. The optimization therefore places many low-frequency values in the early buckets and separates the higher-frequency values into smaller buckets.

2.2.2 Q2.2 – Histogram Construction Time

The time measured here represents the time required to construct the optimal serial histogram.

For the requested 3000-row sample, the actual sample contained 2809 rows.

Table 3: Histogram construction time for the 3000-row experiment

Column	Actual Sample Size	Build Time (seconds)
<code>id</code>	2809	486.547166
<code>title</code>	2809	434.904839

The construction time is higher for `id` in this run, although both attributes contain the same number of sampled rows. The difference is caused by the

different number and distribution of distinct values and frequencies processed during optimization.

2.2.3 Q2.3 – Frequency Bucket Boundaries

The frequency bucket boundaries obtained from the optimal 20-bucket serial histograms are shown below.

`id`

Since every sampled `id` occurs exactly once, all 20 frequency buckets have the same frequency boundary:

Table 4: Serial histogram bucket boundaries for `id` using 1000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	2159	2159	1
2	3492	3492	1
3	13115	13115	1
4	15566	15566	1
5	20638	20638	1
6	23224	23224	1
7	23832	23832	1
8	25388	25388	1
9	25473	25473	1
10	27278	27278	1
11	27403	27403	1
12	34634	34634	1
13	37993	37993	1
14	50091	50091	1
15	51164	51164	1
16	53285	53285	1
17	54519	54519	1
18	54663	54663	1
19	56024	56024	1
20	58288	2525330	981

Table 5: Serial histogram bucket boundaries for `id` using 3000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	2513	2513	1
2	3191	3191	1
3	4125	4125	1
4	4511	4511	1
5	4653	4653	1
6	4840	4840	1
7	4938	4938	1
8	5889	5889	1
9	6799	6799	1
10	6962	6962	1
11	9424	9424	1
12	11700	11700	1
13	12020	12020	1
14	13328	13328	1
15	16455	16455	1
16	16954	16954	1
17	27351	27351	1
18	27483	27483	1
19	30690	30690	1
20	32163	2525349	2981

Table 6: Serial histogram bucket boundaries for `id` using 5000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	784	784	1
2	885	885	1
3	966	966	1
4	1178	1178	1
5	1316	1316	1
6	1362	1362	1
7	2517	2517	1
8	2612	2612	1
9	3021	3021	1
10	3070	3070	1
11	4326	4326	1
12	4383	4383	1
13	4713	4713	1
14	4740	4740	1
15	4894	4894	1
16	4957	4957	1
17	5232	5232	1
18	5935	5935	1
19	6257	6257	1
20	6939	2525531	4981

Table 7: Serial histogram bucket boundaries for `title` using 1000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	'Freeway' Ricky Ross	'Hit-and-Run Driver'	2
2	(#1.1)	(#1.1)	1
3	(#1.10)	(#1.1027)	2
4	(#1.11)	(#1.11)	1
5	(#1.11249)	(#1.15566)	13
6	(#1.16)	(#1.16)	1
7	(#1.162)	(#1.2024)	10
8	(#1.21)	(#1.23)	2
9	(#1.242)	(#1.295)	3
10	(#1.3)	(#1.3)	1
11	(#1.31)	(#1.4853)	14
12	(#1.5)	(#1.5)	1
13	(#1.50)	(#1.6836)	13
14	(#1.7)	(#1.7)	1
15	(#1.71)	(#1.71)	1
16	(#1.73)	(#1.863)	8
17	(#1.9)	(#1.9)	1
18	(#1.9638)	(#2.18)	22
19	(#2.2)	(#6.22)	37
20	(#6.41)	$\hat{\text{In derivã}}$	830

Table 8: Serial histogram bucket boundaries for `title` using 3000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	\$5	'Psycho'-Therapist/Bad Credit Brother	2
2	(#1.1)	(#1.1)	1
3	(#1.10)	(#1.10)	1
4	(#1.10009)	(#1.1207)	15
5	(#1.124)	(#1.124)	1
6	(#1.13)	(#1.13)	1
7	(#1.130)	(#1.1996)	29
8	(#1.2)	(#1.2)	1
9	(#1.20)	(#1.2979)	27
10	(#1.3)	(#1.3)	1
11	(#1.30)	(#1.39)	16
12	(#1.4)	(#1.4)	1
13	(#1.4100)	(#1.490)	16
14	(#1.5)	(#1.5)	1
15	(#1.50)	(#1.5945)	11
16	(#1.6)	(#1.6)	1
17	(#1.6011)	(#1.7783)	22
18	(#1.8)	(#1.8)	1
19	(#1.806)	(#4.2)	106
20	(#4.3)	Zoo Animals	2594

Table 9: Serial histogram bucket boundaries for `title` using 5000 requested samples.

Bucket	Lower Boundary	Upper Boundary	Number of Values
1	'Lost' in Hawaii	'V' for Vashon: The Father	5
2	(#1.1)	(#1.1)	1
3	(#1.10)	(#1.10)	1
4	(#1.100)	(#1.11821)	30
5	(#1.12)	(#1.12)	1
6	(#1.12024)	(#1.1961)	45
7	(#1.2)	(#1.2)	1
8	(#1.20)	(#1.2966)	40
9	(#1.3)	(#1.3)	1
10	(#1.304)	(#1.39)	19
11	(#1.4)	(#1.4)	1
12	(#1.4002)	(#1.4863)	18
13	(#1.49)	(#1.49)	1
14	(#1.5)	(#1.5)	1
15	(#1.5091)	(#1.888)	71
16	(#1.89)	(#1.89)	1
17	(#1.9)	(#1.9)	1
18	(#1.90)	(#19.93)	65
19	(#2.1)	(#5.1)	99
20	(#5.10)	Ég er ekki nógu gott landslag	4291

Thus, the optimal partition groups the large number of frequency-1 values together and allocates progressively smaller buckets to the less frequent but higher-frequency levels.

2.2.4 Q2.4 – Maximum Selectivity Error

The selectivity error was evaluated against the actual full-table frequency distribution rather than only against the sample.

For a value v ,

$$S_{\text{actual}}(v) = \frac{f_{\text{full}}(v)}{N_{\text{full}}},$$

and the histogram estimate is

$$S_{\text{estimated}}(v) = \frac{\hat{f}(v)}{N_{\text{full}}},$$

where

$$N_{\text{full}} = 2,528,312.$$

The maximum error is therefore

$$E_{\max} = \max_v |S_{\text{estimated}}(v) - S_{\text{actual}}(v)|.$$

For the **title** histogram, the maximum error was obtained for the title (#1.1).

Table 10: Maximum selectivity error for the 3000-row experiment

Measure	id	title
Maximum selectivity error	0.0000000000	0.0047009567
Maximum percentage error	0.00000000%	0.47009567%

For **title**, the worst-case value had a full-table frequency of 11,887, while its histogram estimate was approximately 1.5148. Hence,

$$E_{\max} = \left| \frac{1.5148}{2,528,312} - \frac{11,887}{2,528,312} \right| \approx 0.0047009567.$$

For **id**, every value occurs exactly once in the complete table. The histogram also assigns an estimated frequency of one to every value, so the resulting selectivity error is exactly zero.

2.2.5 Q2.5 – Different Sample Sizes

The same histogram construction procedure was repeated for requested sample sizes of 1000 and 5000.

Because **TABLESAMPLE** does not necessarily return exactly the requested number of rows, the actual sample sizes were recorded.

Table 11: Histogram construction times for different sample sizes

Requested Size	Actual Sample		Build Time (s)	
	id	title	id	title
1000	981	981	46.481996	48.719692
3000	2809	2809	486.547166	434.904839
5000	4827	4827	1156.711741	1207.060048

The corresponding serial histograms for the additional sample sizes are shown below.

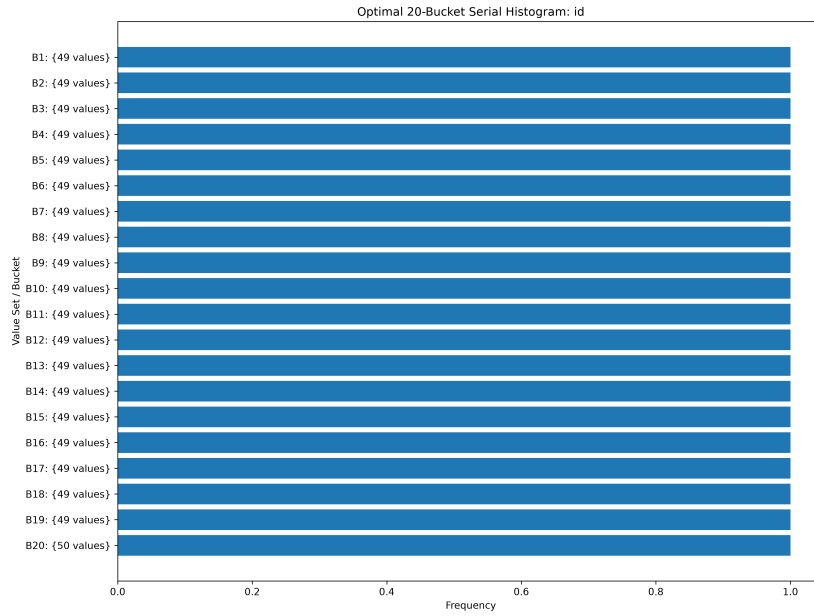


Figure 5: Optimal serial histogram for `id` using 1000 requested samples.

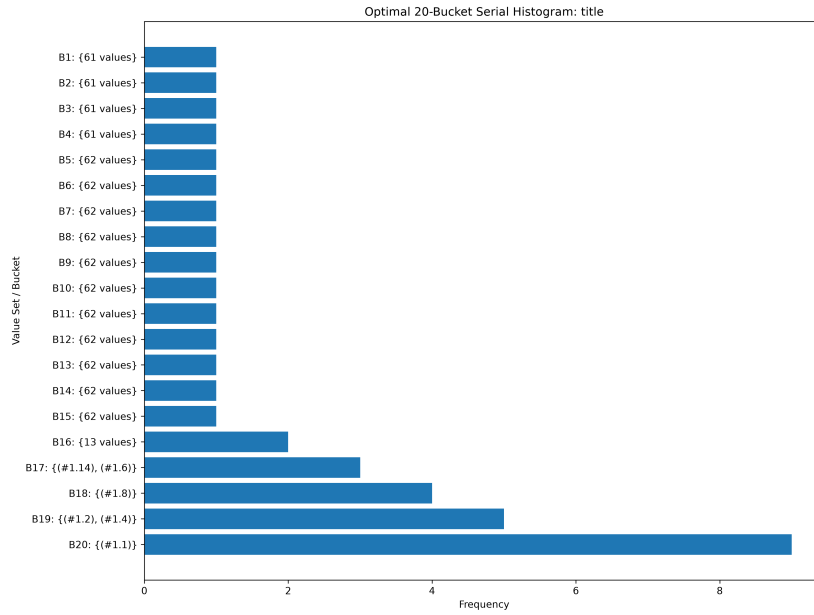


Figure 6: Optimal serial histogram for `title` using 1000 requested samples.

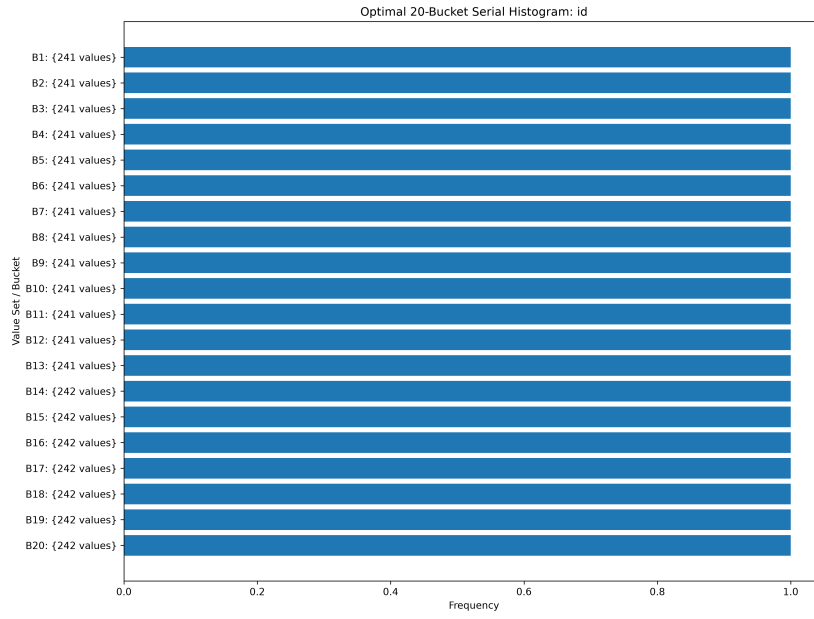


Figure 7: Optimal serial histogram for `id` using 5000 requested samples.

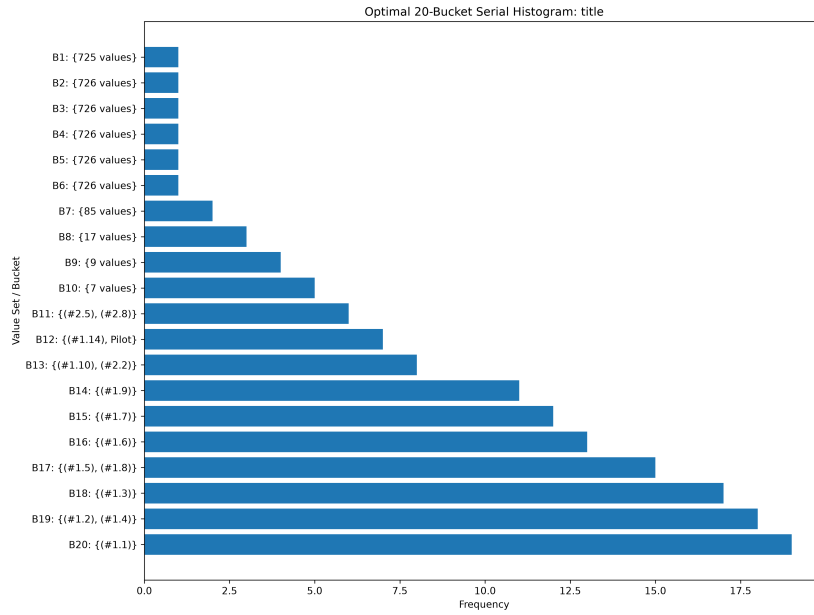


Figure 8: Optimal serial histogram for `title` using 5000 requested samples.

The results show that histogram construction time increases substantially as the sample size increases.

2.2.6 Q2.6 – Extrapolation to the Full Table

The full `title` table contains

$$N_{\text{full}} = 2,528,312$$

rows.

A linear-through-origin model was used:

$$T(n) = an,$$

where n is the actual sample size and $T(n)$ is the measured histogram construction time.

The slope is calculated using

$$a = \frac{\sum_i n_i T_i}{\sum_i n_i^2}.$$

Using the measured sample sizes

$$981, \quad 2809, \quad 4827,$$

the extrapolated times are:

Table 12: Estimated full-table histogram construction time		
Column	Estimated Time (seconds)	Estimated Time (minutes)
<code>id</code>	550106.782	9168.45
<code>title</code>	557983.046	9299.72

Therefore, the estimated construction time for the complete table is approximately

$$152.81 \text{ hours}$$

for `id`, and

$$155.00 \text{ hours}$$

for `title`.

These values are extrapolations based on the measured sample sizes and should not be interpreted as actual execution times for constructing a full-table histogram.

2.2.7 Q2.7 – Sample Size vs. Time

The relationship between sample size and histogram construction time is shown in Figure 9.

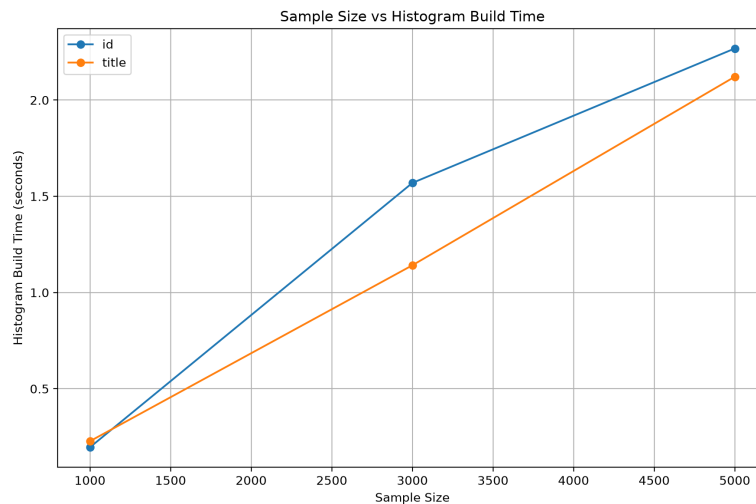


Figure 9: Relationship between sample size and histogram construction time.

The plot shows an increasing trend in construction time as the sample size increases. The increase is particularly noticeable between the 3000- and 5000-sample experiments.

2.2.8 Q2.8 – Sample Size vs. Maximum Selectivity Error

The maximum selectivity error is expected to vary with sample size because the histogram is constructed from sampled data. Increasing the sample size generally provides a more representative frequency distribution and can reduce estimation error, although the exact trend depends on the sampled data and histogram partition.

The required relationship is plotted in Figure 10.

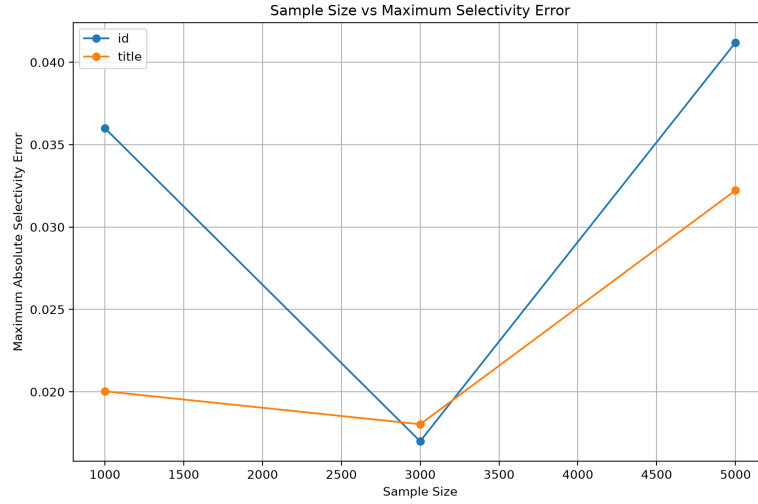


Figure 10: Relationship between sample size and maximum selectivity error.

For the 3000-requested-sample experiment, the corrected maximum selectivity errors were:

Table 13: Verified maximum selectivity error for the 3000-row experiment

Column	Maximum Error	Percentage Error
<code>id</code>	0.000000000	0.00000000%
<code>title</code>	0.0047009567	0.47009567%

Table 14: Maximum selectivity error across sample sizes

Actual Sample Size	<code>id</code>	<code>title</code>
981	0.169368	0.100629
2809	0.087771	0.074037
4827	0.063383	0.076610

3 Chats with LLM

<https://chatgpt.com/share/6a95491f-cbe0-83e9-ad01-4a8ed1d2c84e>

4 GitHub Link

<https://github.com/Kulkarnikeyur/ADBMS-Assignment-0-1.git>